



Module 1 : Time and work (2 marks)

1. If Roger can do a piece of work in 8 days and Antony can complete the same work in 5 days, in how many days will both of them together complete it?

$$\text{Roger's 1 day's work} = \frac{1}{8}; \text{Antony's 1 day's work} = \frac{1}{5}.$$

$$(\text{Roger} + \text{Antony})\text{'s 1 day's work} = \left(\frac{1}{8} + \frac{1}{5}\right) = \frac{13}{40}.$$

$$\therefore \text{Both Roger and Antony will complete the work in } \frac{40}{13} = 3\frac{1}{13} \text{ days.}$$

2. A alone can complete a piece of work of Rs. 300 in 6 days; but by engaging an assistant, the work is completed in 4 days. Find the share to be received by the assistant.

$$\text{Assistant's 1 day's work} = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

$$\therefore \text{A's share : Assistant's share} = \text{Ratio of their 1 day's work} = \frac{1}{6} : \frac{1}{12} = 2 : 1.$$

$$\text{Hence, assistant's share} = ₹ \left(300 \times \frac{1}{3} \right) = ₹ 100.$$

3. A can do a work in 12 days and B in 18 days. How long will they take together?

Total work = 36 unit

$$1 \text{ day work of A} = 36/12 = 3 \text{ unit}$$

$$1 \text{ day work of B} = 36/18 = 2 \text{ unit}$$

B alone worked for 3 days, (since As A left the work 3 days before so, B has to do the left work alone for 3 days.)

$$\Rightarrow \text{Total work done by B alone in 3 days} = 3 \times 2 \text{ unit} = 6 \text{ unit work,}$$

$$\Rightarrow \text{Total work left} = 36 \text{ unit} - 6 \text{ unit} = 30 \text{ unit.}$$

$$\Rightarrow 30 \text{ units work is done by both A and B}$$

$\therefore$ Number of days to complete the remaining work = 30/total work done by A and B in 1 day

$$\Rightarrow 30/5 = 6 \text{ days}$$

4. A pipe fills a tank in 6 hours. Another empties it in 12 hours. Total time to fill?



$$\text{Filling rate} = \frac{1}{6} \text{ tank/hour}$$

$$\text{Emptying rate} = \frac{1}{12} \text{ tank/hour}$$

$$\Rightarrow \text{Net rate} = \frac{1}{6} - \frac{1}{12} = \frac{2-1}{12} = \frac{1}{12} \text{ tank/hour}$$

$$\text{Required work} = \frac{2}{3} \text{ tank}$$

$$\Rightarrow \text{Time} = \text{Required work} \div \text{Net rate}$$

$$\Rightarrow \text{Time} = \frac{2}{3} \div \frac{1}{12} = \frac{2}{3} \times 12 = 8 \text{ hours}$$

The tank will be two-third filled in 8 hours.

5. 8 men can do a job in 10 days. How long will 4 men take?

$$\begin{aligned} M_1 &= 8 & M_2 &= 4 \\ D_1 &= 10 & D_2 &= ? \\ \frac{M_1 D_1 H_1}{W_1} &= \frac{M_2 D_2 H_2}{W_2} \\ 8 \times 10 &= 4 \times D_2 \\ D_2 &= \frac{8 \times 10}{4} = 20 \text{ days} \end{aligned}$$

6. 15 workers make 300 items in 5 days. How many items will 5 workers make in 5 days?

First find items per worker per day:

$$\text{Items per worker per day} = \frac{300}{15 \times 5} = \frac{300}{75} = 4$$

Now for 5 workers in 5 days:

$$\text{Items} = 5 \times 5 \times 4 = 100$$

Module 1 : Time and work (5 marks)

7. A can do a piece of work in 10 days and B in 20 days. They work together but 2 days before the completion of the work, A leaves. In how many days was the work completed ?



$$B's\ 2\ days'\ work = \left(\frac{1}{20} \times 2\right) = \frac{1}{10}. \text{ Remaining work} = \left(1 - \frac{1}{10}\right) = \frac{9}{10}.$$

$$(A + B)'s\ 1\ day's\ work = \left(\frac{1}{10} + \frac{1}{20}\right) = \frac{6}{40} = \frac{3}{20}.$$

Now, $\frac{3}{20}$ work is done by A and B in 1 day.

$$\therefore \frac{9}{10} \text{ work is done by A and B in } \left(\frac{20}{3} \times \frac{9}{10}\right) = 6 \text{ days.}$$

Hence, total time taken = (2 + 6) days = 8 days.

8. A tap can fill a tank in 48 minutes whereas another tap can empty it in 2 hours. If both the taps are opened at 11 : 40 A.M., then the tank will be filled at what time?

$$\text{Net part filled in 1 hour} = \left(\frac{1}{48} - \frac{1}{120}\right) = \frac{3}{240} = \frac{1}{80}.$$

$\therefore$ The tank will be filled 80 mins i.e. 1 hour 20 min.
after 11 : 40 A.M. i.e. at 1 P.M.

9. A and B undertake to do a piece of work for Rs. 600. A alone can do it in 6 days while B alone can do it in 8 days. With the help of C, they finish it in 3 days. Find the share of each.

$$C's\ 1\ day's\ work = \frac{1}{3} - \left(\frac{1}{6} + \frac{1}{8}\right) = \frac{1}{24}.$$

$$\therefore A : B : C = \text{Ratio of their 1 day's work } \frac{1}{6} : \frac{1}{8} : \frac{1}{24} = 4 : 3 : 1.$$

$$\therefore A's\ share = ₹ \left(600 \times \frac{4}{8}\right) = ₹ 300, \ B's\ share = ₹ \left(600 \times \frac{3}{8}\right) = ₹ 225.$$

$$C's\ share = ₹ [600 - (300 + 225)] = ₹ 75.$$

10. A and B can do a piece of work in 9 days; B and C can do it in 12 days; A and C can do it in 18 days. In how many days will A, B and C finish it, working together and separately?



$$(A + B)\text{'s 1 day's work} = \frac{1}{9}; (B + C)\text{'s 1 day's work} = \frac{1}{12};$$

$$(A + C)\text{'s 1 day's work} = \frac{1}{18}.$$

$$\text{Adding, we get: } 2(A + B + C)\text{'s 1 day's work} = \left(\frac{1}{9} + \frac{1}{12} + \frac{1}{18}\right) = \frac{9}{36} = \frac{1}{4}.$$

$$\therefore (A + B + C)\text{'s 1 day's work} = \frac{1}{8}.$$

Thus, A, B and C together can finish the work in 8 days.

Now, A's 1 day's work = [(A + B + C)'s 1 day's work - (B + C)'s 1 day's work]

$$= \left(\frac{1}{8} - \frac{1}{12}\right) = \frac{1}{24}.$$

$\therefore$ A alone can finish the work in 24 days.

Similarly, B's 1 day's work

$$= [(A + B + C)\text{'s 1 day's work} - (A + C)\text{'s 1 day's work}] = \left(\frac{1}{8} - \frac{1}{18}\right) = \frac{5}{72}.$$

$\therefore$ B alone can finish the work in $\frac{72}{5} = 14\frac{2}{5}$ days.

And, C's 1 day's work

$$= [(A + B + C)\text{'s 1 day's work} - (A + B)\text{'s 1 day's work}] = \left(\frac{1}{8} - \frac{1}{9}\right) = \frac{1}{72}.$$

$\therefore$ C alone can finish the work in 72 days.



11. A can do a piece of work in 80 days. He works at it for 10 days and then B alone finishes the remaining work in 42 days. In how much time will A and B, working together, finish the work ?

$$\text{Work done by A in 10 days} = \left(\frac{1}{80} \times 10 \right) = \frac{1}{8}. \text{ Remaining work} = \left(1 - \frac{1}{8} \right) = \frac{7}{8}$$

Now, $\frac{7}{8}$ work is done by B in 42 days.

$$\text{Whole work will be done by B in } \left(42 \times \frac{8}{7} \right) = 48 \text{ days.}$$

$$\therefore \text{A's 1 day's work} = \frac{1}{80} \text{ and B's 1 day's work} = \frac{1}{48}.$$

$$\therefore (\text{A} + \text{B})\text{'s 1 day's work} = \left(\frac{1}{80} + \frac{1}{48} \right) = \frac{8}{240} = \frac{1}{30}.$$

Hence, both will finish the work in 30 days.

12. 2 men and 3 boys can do a piece of work in 10 days while 3 men and 2 boys can do the same work in 8 days. In how many days can 2 men and 1 boy do the work?

Let 1 man's 1 day's work = x and 1 boy's 1 day's work = y .

$$\text{Then, } 2x + 3y = \frac{1}{10} \text{ and } 3x + 2y = \frac{1}{8}.$$

$$\text{Solving, we get : } x = \frac{7}{200} \text{ and } y = \frac{1}{100}.$$

$$\therefore (2 \text{ men} + 1 \text{ boy})\text{'s 1 day's work} = \left(2 \times \frac{7}{200} + 1 \times \frac{1}{100} \right) = \frac{16}{200} = \frac{2}{25}.$$

$$\text{So, 2 men and 1 boy together can finish the work in } \frac{25}{2} = 12 \frac{1}{2} \text{ days.}$$

Module 2 : PnC & Probability (2 marks)

13. In how many different ways can the letters of the word 'FIGHT' be arranged?

The given word contains 5 different letters.

$$\text{Required number of ways} = {}^5P_5 = \underline{5!} = (5 \times 4 \times 3 \times 2 \times 1) = 120.$$



14. In how many different ways can the letters of the word 'PRESENT' be arranged?

The word 'PRESENT' contains 7 letters, namely 2E and all other 5 are different.

$$\therefore \text{ Required number of ways} = \frac{7!}{2!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2} = 2520.$$

15. Evaluate : (i) 5P_5 (ii) ${}^{50}P_3$

$$(i) {}^5P_5 = \frac{5!}{(5-5)!} = 5! = (5 \times 4 \times 3 \times 2 \times 1) = 120.$$

$$(ii) {}^{50}P_3 = \frac{50!}{(50-3)!} = \frac{50!}{47!} = \frac{50 \times 49 \times 48 \times (47!)}{47!} = (50 \times 49 \times 48) = 117600.$$

16. In a throw of a coin, find the probability of getting a head.

Here $S = \{H, T\}$ and $E = \{H\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{1}{2}.$$

17. Two unbiased coins are tossed. What is the probability of getting at most one head ?

Here $S = \{HH, HT, TH, TT\}$.

Let $E =$ event of getting at most one head.

$\therefore E = \{TT, HT, TH\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}.$$

18. An unbiased die is tossed. Find the probability of getting a multiple of 3.

Here $S = \{1, 2, 3, 4, 5, 6\}$.

Let E be the event of getting a multiple of 3.

Then, $E = \{3, 6\}$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{2}{6} = \frac{1}{3}.$$

Module 2 : PnC & Probability (5 marks)

19. How many arrangements can be made out of the letters of the word 'ENGINEERING'?

The word 'ENGINEERING' contains 11 letters, namely 3E, 3N, 2G, 2I and 1R.

$$\begin{aligned} \therefore \text{ Required number of ways} &= \frac{11!}{(3!)(3!)(2!)(2!)(1!)} \\ &= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 6 \times 2 \times 2 \times 1} \\ &= (11 \times 10 \times 7 \times 6 \times 5 \times 4 \times 3) = 277200. \end{aligned}$$



20. In how many ways, a committee of 6 members be selected from 7 men and 5 ladies, consisting of 4 men and 2 ladies?

We have to select (4 men out of 7) and (2 ladies out of 5).

$$\therefore \text{Required number of ways} = {}^7C_4 \times {}^5C_2 = {}^7C_3 \times {}^5C_2 = \left(\frac{7 \times 6 \times 5}{3 \times 2 \times 1} \times \frac{5 \times 4}{2 \times 1} \right) = 350.$$

21. Out of 5 women and 4 men, a committee of three members is to be formed in such a way that at least one member is a woman. In how many different ways can it be done?

$$\text{Required number of ways} = ({}^5C_1 \times {}^4C_2) + ({}^5C_2 \times {}^4C_1) + ({}^5C_3)$$

$$= \left(5 \times \frac{4 \times 3}{2 \times 1} \right) + \left(\frac{5 \times 4}{2 \times 1} \times 4 \right) + \left(\frac{5 \times 4 \times 3}{3 \times 2 \times 1} \right)$$

$$= (30 + 40 + 10) = 80.$$

22. In how many different ways can the letters of the word 'TRANSPIRATION' be arranged so that the vowels always come together?

The word TRANSPIRATION has 13 letters in which each of T, R, A, N and I has come two time

We have to arrange TT RR NN PS (AA II O).

There are five vowels in the given words.

$\therefore$ we consider these five vowels as one letter.

$\therefore$ Required number of arrangements.

$$= \frac{9! \times 5!}{2!2!2!2!2!}$$

$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 5 \times 4 \times 3 \times 2}{2 \times 2 \times 2 \times 2 \times 2}$$

$$= 1360800$$

23. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white ?

$$\text{Total number of balls} = (4 + 5 + 6) = 15.$$

$$P(\text{drawing a red ball or a green ball}) = P(\text{red}) + P(\text{green})$$

$$= \left(\frac{4}{15} + \frac{5}{15} \right) = \frac{9}{15} = \frac{3}{5}.$$

24. One card is drawn from a pack of 52 cards. What is the probability that the card drawn is either a red card or a king ?



Here, $n(S) = 52$.

There are 26 red cards (including 2 kings) and there are 2 more kings.

Let E = event of getting a red card or a king.

Then, $n(E) = 28$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{28}{52} = \frac{7}{13}.$$

Module 3 : clocks and calendars (2 marks)

25. (i) How many degrees are there in a full clock? Ans : 360

(ii) What is the angle between two consecutive numbers on a clock? Ans: 30

26. (i) What is the angle between the hour and minute hands at 3:00? Ans: 90

(ii) What is the angle between the hands at 6:00? Ans: 180

27. At what time between 2 and 3 o'clock will the hands of a clock be together ?

At 2 o'clock, the hour hand is at 2 and the minute hand is at 12, i.e. they are 10 min. spaces apart. To be together, the minute hand must gain 10 minutes over the hour hand.

Now, 55 minutes are gained by it in 60 min.

$$\therefore 10 \text{ minutes will be gained in } \left(\frac{60}{55} \times 10 \right) \text{ min.} = 10 \frac{10}{11} \text{ min.}$$

$$\therefore \text{The hands will coincide at } 10 \frac{10}{11} \text{ min. past 2.}$$

Module 3 : clocks and calendars (5 marks)

28. The angle between the minute hand and the hour hand of a clock when the time (i) 8.30 (ii) 4.20

Angle traced by hour hand in

$$\frac{17}{2} \text{ hrs} = \left(\frac{360}{12} \times \frac{17}{2} \right)^\circ = 255^\circ.$$

$$\text{Angle traced by min. hand in 30 min.} = \left(\frac{360}{60} \times 30 \right)^\circ = 180^\circ.$$

$$\therefore \text{Required angle} = (255 - 180)^\circ = 75^\circ.$$

Angle traced by hour hand in

$$\frac{13}{3} \text{ hrs} = \left(\frac{360}{12} \times \frac{13}{3} \right)^\circ = 130^\circ.$$

$$\text{Angle traced by min. hand in 20 min.} = \left(\frac{360}{60} \times 20 \right)^\circ = 120^\circ.$$

$$\therefore \text{Required angle} = (130 - 120)^\circ = 10^\circ.$$



29. At what time between 6 and 7 o'clock are the hands of a clock coincide to each other?

$$\begin{aligned} M &= \frac{2}{11} (30H \pm \theta) \\ &= \frac{2}{11} (30(6) \pm 0) \\ &= \frac{2}{11} (180) = \frac{360}{11} \\ M &= 32\frac{8}{11} \\ \text{Coincide at } 6:32\frac{8}{11} \end{aligned}$$

coincide = 0
 $\theta = 0^\circ$
6 and 7 o'clock
 $H = 6$

$\theta = (30H \pm$

30. (a) Find the angle between the hour hand and the minute hand at 6:20.

$$\begin{aligned} &6:20 \\ &H \quad M \\ &H=6 \quad M=20 \\ \theta &= 30H - \frac{11}{2}M \\ &= 30(6) - \frac{11}{2}(20) \\ &= 180 - 110 \\ \theta &= 70^\circ \end{aligned}$$

(b) At what time between 5 and 6 o'clock are the hands of a clock right angle to each other?

$$\begin{aligned} &5 \text{ and } 6 \\ &H \quad \theta = 90^\circ \\ M &= \frac{2}{11} (30H \pm \theta) \\ &= \frac{2}{11} (30(5) \pm 90) \\ &= \frac{2}{11} (150 \pm 90) \\ &= \frac{2}{11} (240) \quad \frac{2}{11} (60) \\ &= \frac{480}{11} \quad = \frac{120}{11} \\ &= 43\frac{7}{11} \text{ min} \quad = 10\frac{10}{11} \text{ min} \\ &5:43\frac{7}{11} \text{ min}, \quad 5:10\frac{10}{11} \text{ min} \end{aligned}$$



Objective- (Fill in the Blanks)

1. Number of all permutations of n things, taking all at a time is $n!$
2. AB and BA represent the same combination.
3. In 364 many ways can a cricket eleven be chosen out of 14 players?
4. AB and BA are two different permutations.
5. 5040 different ways can the letters of the word DISPLAY be arranged?
6. Three balls are drawn from a bag containing 5 red, 4 white and 3 black balls. The number of ways in which this can be done if at least 2 are red is 80
7. If ${}^{12}C_5 = {}^{12}C_r$ then $r =$ 5 or 7.
8. $7! =$ 5040
9. ${}^5P_3 + {}^5C_3 =$ $60 + 10 = 70$
10. $6! + 5! =$ 840
11. A can complete a work in 10 days. A's one day work = $1/10$
12. If a man can do a work in 25 days, then work done in 1 day =
Answer: $1/25$
13. If A can do a work in 20 days, then A's one day work =
Answer: $1/20$
14. If a work is completed in 5 days, then total work is taken as
Answer: 1 unit
15. If A does a work in 10 days and B in 10 days, together they take days
Answer: 5 days
16. If a man works for 3 days out of 12 days, work done =
Answer: $1/4$
17. If two men do equal work, their wages are
Answer: equal
18. Wages are directly proportional to the amount of done
Answer: work
19. If A and B earn Rs. 300 in total and A gets Rs.200, B gets Rs
Answer: 100



20. If A works twice as much as B, A's wages will be _____ **times** B's wages
Answer: 2
21. If 5 men earn Rs. 500 in one day, then 1 man earns Rs _____
Answer: 100
22. If number of workers increases, time taken to do the same work _____
Answer: decreases
23. If 10 men can do a work in 6 days, then 1 man will take _____ **days**
Answer: 60 days
24. Men and days are _____ **proportional** to each other
Answer: inversely
25. If working hours increase, total days required _____
Answer: decrease
26. If 4 men can do a work in 8 days, then 8 men can do it in _____ **days**
Answer: 4 days
27. If a pipe can fill a tank in 10 hours, part filled in 1 hour = _____
Answer: 1/10
28. If one pipe fills a tank in 12 hours, it will fill _____ part of the tank in 1 hour
Answer: 1/12
29. If a pipe fills a tank in 8 hours, it fills the whole tank in _____ hours
Answer: 8
30. If two pipes together fill a tank in 4 hours, part filled in 1 hour = _____
Answer: 1/4

Objective (Multiple choice questions)

10 x (1/2) =5 Marks

1. A man can do a job in 15 days. His father takes 20 days and his son finishes it in 25 days. How long will they take to complete the job if they all work together? { C }
(a) Less than 6 days (b) Exactly 6 days (c) Approximately 6.4 days (d) 10 days
2. Amit and Sumit can plough a field in 4 days. Sumit alone can plough the field in 6 days. In how many days will Amit alone plough the field? { b }
(a) 10 days (b) 12 days (c) 14 days (d) 15 days
3. A can do a piece of work in 8 days and B can do the same piece of work in 12 days. A and B together complete the same piece of work and get Rs.200 as the combined wages. B's share of the wages will be (in Rs) { b }
(a) 75 (b) 80 (c) 85 (d) 90



4. Reena, Aastha and Shloka can independently complete a piece of work in 6 hours, 4 hours and 12 hours respectively. If they work together, how much time will they take to complete that piece of work? { a }
(a) 2 hours (b) 5 hours (c) 6 hours (d) 8 hours
5. X, Y and Z complete a work in 6 days. X or Y alone can do the same work in 16 days. In how many days Z alone can finish the same work? { c }
(a) 12 (b) 16 (c) 24 (d) 36
6. X can do $\frac{1}{4}$ of a work in 10 days, Y can do 40% of the work in 40 days and Z can do $\frac{1}{3}$ of the work in 13 days. Who will complete the work first ? { c }
(a) X (b) Y (c) Z (d) X and Z both
7. A man and a boy together can do a certain amount of digging in 40 days. Their speeds in digging are in the ratio of 8 : 5. How many days will the boy take to complete the work if engaged alone? { d }
(a) 52 days (b) 68 days (c) 80 days (d) 104 days
8. A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in : { b }
(a) 4 days (b) 6 days (c) 8 days (d) 12 days
9. Rosa can eat 32 rosogollas in one hour. Her sister Lila needs three hours to eat the same number. How much time will they take to eat 32 rosogollas together? { a }
(a) 45 minutes (b) 75 minutes (c) 90 minutes (d) None of these
10. A and B can do a piece of work in 5 days; B and C can do it in 7 days; A and C can do it in 4 days. Who among these will take the least time if put to do it alone ? { a }
(a) A (b) B (c) C (d) Data inadequate
11. How many ways can you arrange 3 letters A, B, C?
a) 3
b) 6
c) 9
d) 12
Answer: b) 6
(Reason: $3! = 6$)
12. How many ways can 4 people sit in a row?
a) 4



- b) 8
- c) 16
- d) 24

Answer: d) 24

(Reason: $4! = 24$)

13. How many ways can you select 2 fruits from 3 fruits (Apple, Banana, Mango)?

- a) 2
- b) 3
- c) 4
- d) 6

Answer: b) 3

(Reason: Combination ${}^3C_2 = 3$)

14. Number of ways to arrange 4 books in a **circle**?

- a) 6
- b) 12
- c) 24
- d) 16

Answer: a) 6

(Reason: Circular permutation formula $= (n-1)! = 3! = 6$)

15. **Q5.** How many ways can 3 students be selected from 5 students?

- a) 5
- b) 10
- c) 15
- d) 20

Answer: b) 10

(Reason: ${}^5C_3 = 10$)

16. A coin is tossed. Probability of getting **heads** = ?

- a) 1
- b) $1/2$
- c) $1/4$
- d) 2

Answer: b) $1/2$

17. A die is rolled. Probability of getting **4** = ?

- a) $1/2$
- b) $1/3$
- c) $1/6$
- d) $1/4$

Answer: c) $1/6$

18. A card is drawn from a standard deck of 52 cards. Probability of getting a **king** = ?

- a) $1/52$
- b) $1/13$
- c) $4/52$
- d) Both b & c

Answer: d) Both b & c

(Reason: $4 \text{ kings} / 52 \text{ cards} = 1/13$)



19. A die is rolled. Probability of getting **even number** = ?

- a) $1/2$
- b) $1/3$
- c) $1/6$
- d) $2/3$

Answer: a) $1/2$

(Reason: $2, 4, 6 \rightarrow 3 \text{ outcomes} / 6 \text{ total} = 1/2$)

20. Two coins are tossed. Probability of getting **one head and one tail** = ?

- a) $1/4$
- b) $1/2$
- c) $1/3$
- d) $3/4$

Answer: b) $1/2$

(Reason: Outcomes: $HH, HT, TH, TT \rightarrow 2 \text{ favorable} / 4 = 1/2$)

21. **Q1.** How many ways can the letters of the word **CAT** be arranged?

- a) 3
- b) 6
- c) 9
- d) 12

Answer: b) 6

22. How many ways can 5 people sit in a circle?

- a) 120
- b) 24
- c) 1200
- d) 24

Answer: b) 24

(Circular permutation: $(n-1)! = 4! = 24$)

23. How many ways can 3 prizes be given to 5 students if each student can get only one prize?

- a) 10
- b) 60
- c) 15
- d) 20

Answer: b) 60

(Permutation: $5P3 = 5 \times 4 \times 3 = 60$)

24. In how many ways can the first 3 ranks be assigned to 7 students?

- a) 35
- b) 210
- c) 21
- d) 7

Answer: b) 210

(Permutation: $7P3 = 7 \times 6 \times 5 = 210$)



25. How many ways can 4 books be arranged on a shelf?

- a) 4
- b) 8
- c) 16
- d) 24

Answer: d) 24

(Permutation: $4! = 24$)

26. How many ways can 2 letters be selected from 4 letters?

- a) 4
- b) 6
- c) 8
- d) 12

Answer: b) 6

(Combination: $4C2 = 6$)

27. How many ways can 3 students sit on a circular table?

- a) 6
- b) 3
- c) 2
- d) 1

Answer: a) $2! = 2$ (Wait – careful!)

28. Check carefully: Circular permutation of n items = $(n-1)!$

3 students $\rightarrow (3-1)! = 2$

✓ **Answer: 2**

29. In a competition, 1st, 2nd, 3rd prizes are awarded among 4 contestants. How many arrangements?

- a) 12
- b) 24
- c) 16
- d) 4

Answer: a) 24

($4P3 = 4 \times 3 \times 2 = 24$)

30. A die is rolled. Probability of getting a number greater than 4 = ?

- a) $1/2$
- b) $1/3$
- c) $1/6$
- d) $2/3$

Answer: a) $1/3$

(Numbers 5, 6 $\rightarrow 2/6 = 1/3$)

31. Two coins are tossed. Probability of getting **at least one tail** = ?

- a) $1/2$
- b) $3/4$
- c) $1/4$



d) 1

Answer: b) $\frac{3}{4}$

(Outcomes: HH, HT, TH, TT $\rightarrow$ 3 favorable)

32. One card is drawn from a deck of 52. Probability it is a **red card** = ?

a) $\frac{1}{2}$

b) $\frac{1}{4}$

c) $\frac{1}{3}$

d) $\frac{1}{13}$

Answer: a) $\frac{1}{2}$

(Red cards = $\frac{26}{52}$)

33. A bag contains 5 white and 3 black balls. A ball is drawn randomly. Probability it is **white** = ?

a) $\frac{5}{8}$

b) $\frac{3}{8}$

c) $\frac{1}{2}$

d) $\frac{1}{3}$

Answer: a) $\frac{5}{8}$

34. Two dice are rolled. Probability of getting a sum = 7?

a) $\frac{1}{6}$

b) $\frac{1}{8}$

c) $\frac{1}{12}$

d) $\frac{1}{36}$

Answer: a) $\frac{1}{6}$

(Favorable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) $\rightarrow \frac{6}{36} = \frac{1}{6}$)

35. A bag contains 2 red and 3 green balls. One ball is drawn randomly. Probability it is **green** = ?

a) $\frac{2}{5}$

b) $\frac{3}{5}$

c) $\frac{1}{5}$

d) $\frac{1}{2}$

Answer: b) $\frac{3}{5}$

36. A coin is tossed 3 times. Probability of getting **exactly 2 heads** = ?

a) $\frac{1}{8}$

b) $\frac{1}{2}$

c) $\frac{3}{8}$

d) $\frac{1}{4}$

Answer: c) $\frac{3}{8}$

(HH tails combinations: HHT, HTH, THH $\rightarrow \frac{3}{8}$)